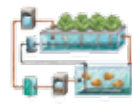




Hydroponics Market Research Report

This research report details the global forecast of hydroponics until the year 2024. <https://digit.in/feb20-05>



Aquaponics!

Learn more about aquaponics, a practice that mixes aquaculture with hydroponics and its uses. <https://digit.in/feb20-06>

If one of you went to your grandfather and told him that people are growing plants and crops solely using water and have eliminated the soil use, he would've probably sighed and thought "Is this why we spend money on this *nalayak's* education?"

However, this is actually a practice. Plants can be and have been grown (for a while now) using almost no soil and "hydroponics" is the name given to this practice. Plants' roots are submerged in a water-based liquid, with extra nutrients and minerals added which are essential for the growth of the plants. In some cases, a medium is used so that the roots can hold the plants' weight and keep them upright. This medium

Hydroponics is not, by any means, a new technology. Let's trace this practice right back to its origins.

THE ORIGINS:

The origins of hydroponics can be traced back to the ancient city of Babylon, where present-day Iraq is situated. The Hanging Gardens of Babylon, one of the Seven Wonders of the Ancient World, is the first known example of soil-less growth of plants. This was around 600 BC. Later on, around 1100 BC, the Aztec Indians got creative with their growing techniques and created gardens that seemed to be floating. These "floating gardens" were called 'chinampas', which had a strong combination of roots and lashes, laden with sediment from lake-bottoms, providing nutrients to the crops and plantations.

tained his famous branching rule: "all the branches of a tree at every stage of its height when put together are equal in thickness to the trunk".

THE JOURNEY:

In 1627, *Sylva Sylvarum* was published. Written by Francis Bacon, *Sylva Sylvarum* contained the earliest work on growing terrestrial plants with soil-less techniques. Major discoveries were made in the 17th century. Jean Baptista van Helmont, a Belgian chemist, discovered and proved that plants obtain the necessary minerals and nutrients (he called them "substances") from water. He performed the willow tree experiment, where the tree gained 164 pounds or 74.4 kg but the soil amount remained the same throughout the experiment. John Woodward

HYDROPONICS

Agriculture technology of the future, borrowed from the past.

Adhyayan Panwar | feedback@digit.in

is porous so that it can hold moisture and oxygen that the plants will require. The most popular media are clay, coir, brick shards, etc. The solution contains all important nutrients present in appropriate proportions which ensure optimal growth. The main constituents are Nitrogen (N), Potassium (K), Calcium (Ca), Magnesium (Mg), etc. These are just some of the sixteen elements that are required by plants. Growing systems that don't use a medium are called liquid hydroponic systems which includes NFT (nutrient film technique, in which plants are put in slits in a tube through which nutrient solution is pumped), floating hydroponics, etc. When a medium is used, it's called aggregate hydroponics system which is further divided into open and closed systems. Rockwool is the most widely used medium in open system hydroponics.

Similar floating plantations were discovered by Marco Polo on his visit to China, which left him bewildered as he hadn't seen anything like this ever before. First scientific perspective towards this soil-less technique is attributed to the Italian genius, Leonardo da Vinci, who observed that plants and crops needed to absorb minerals to survive. His findings were published after his death, that con-



Hydroponics can be traced back to ancient times

performed the spear-mint experiment, which showed that plants grow better in impure water compared to distilled water. The first standardised nutrient solution was created in the 18th century by Julius von Schaps and Wilhelm Knop which laid the foundations of hydroponics as we know it today. This solution made it possible to grow roots without any medium. Knop went on to create yet another solution, which is still in use today.

The word "hydroponic" was first used in 1937, coined by Dr. William Frederick Gericke, aptly known as the "father of hydroponics" (he made news when he was able to grow tomato vines more than 7-meters long just using mineral-nutrients solutions, right in his backyard!). The word is a combination of the Greek words "hydro" (meaning water) and "ponos" (meaning labour). This means water-